Jayaditya Vetsa

Coppell, TX | (737) 230-0202 | jayadityavetsa@utexas.edu | Portfolio | GitHub | LinkedIn

EDUCATION

The University of Texas at Austin Expected May 2029
B.S. in Computer Science | Robotics Honors Program | Minor in Robotics *Austin, TX*
GPA: 4.0/4.0

EXPERIENCE

Undergraduate Research Assistant May 2026 - Present
The University of Texas at Dallas | Advisor: Prof. Vibhav Gogate *Richardson, TX*

- Built a **Python/PyTorch robot-motion evaluation pipeline** with Point4D and CAD-based kinematics; benchmarked **11,264 surface points per video** to diagnose tracking failures and assess training-data quality.
- Reduced position error **21%** and increased tracking within **5 cm from 36% to 51%** across **six synthetic clips** by implementing vision-guided refinement with **PyTorch and OpenCV**.
- Reduced held-out robot-pose error from **19.99 to 2.01 degrees MAE** across **793 frames from 12 unseen episodes**.
- Developing **MimicScene for Franka Panda manipulation in Isaac Sim**; integrated **SE(3) alignment, inverse kinematics, collision checking, grasp optimization, and four imitation-learning policies**.

Machine Learning Intern Summers 2023 - 2025
Calimatic Ed Tech *Atlanta, GA / Remote*

- Deployed a **Python RAG service on Azure with OpenAI and Pinecone**; delivered **1,000+ successful executions**.
- Improved operational efficiency **70%** through knowledge-base optimization; shipped **5+ n8n workflows**.

UT INVOLVEMENT

Texas Aerial Robotics | Raytheon AVC Software Team Sept. 2026 - Present

- Evaluated **5-DOF arm tracking, clearance, and effort** at **1/0.5 ms physics steps** using **Gazebo/ROS 2 simulation**.

Texas Humanoids | State Estimation & Perception Engineer Sept. 2026 - Present

- Preparing **humanoid perception and state-estimation workflows** by configuring **OpenCV, ROS 2, and analysis tools**.

SOFTWARE PROJECTS

Robot SphereShield | *C++17, Python, PyTorch, PyBullet* | GitHub Jul. - Sep. 2026

- Raised collision-classification accuracy from **80.71% to 94.28%** on **20,000 held-out poses** by replacing a neural baseline with a **61-sphere PyTorch/URDF model**; observed **0/2,479 false-safe collision predictions**.
- Implemented a **C++17 analysis CLI**; validated **69,400 sphere-pair clearances across 100 configurations** against a Python reference at **1e-10 m tolerance**, with boundary and differential tests in **CMake/CTest**.

SplineFlow-Panda | *Python, MuJoCo, Control Benchmarking* | GitHub Aug. 2026

- Compared **10 Hz action chunks and continuous B-spline trajectories** using a shared **100 Hz controller** across **eight MuJoCo rollouts**, holding paired layouts and evaluation criteria fixed.
- Completed **2/2 paired 2x-speed trials in 5.00 s** with B-splines while action chunks failed the **2 cm waypoint criterion**; analyzed tracking error, jerk, contacts, and actuator saturation.

AI-GP Autonomous Drone Stack | *Python, OpenCV, MAVLink* | GitHub Aug. 2026 - Present

- Integrated **camera-based PnP pose estimation and Kalman-filtered IMU data** for onboard state estimation; engineered **MAVLink control modes** for detection, alignment, and autonomous navigation.
- Built **UDP camera ingestion and MAVLink control**; logged telemetry and implemented recovery logic for **signal dropout, localization failures, and collision events** during runs.

TECHNICAL SKILLS

Languages: Python, C++17, MATLAB, R, Java, SQL
Robotics and Simulation: ROS 2, Gazebo, Isaac Sim, MuJoCo, PyBullet, URDF, LeRobot
Perception and Controls: OpenCV, PnP pose estimation, Kalman filtering, sensor fusion, MAVLink, inverse kinematics
Machine Learning: PyTorch, JAX, TensorFlow, scikit-learn, imitation learning, VLA models, world models
Data and Evaluation: NumPy, pandas, Matplotlib, Seaborn, held-out evaluation, controlled experiments
Engineering Tools: Linux, Git, Docker, CMake/CTest, GitHub Actions, pytest, Azure